

Rational-point decision problems: a conditional transfer and dependency audit

Alec Kriebel

ORCID: [0009-0001-9320-500X](#)

October 6, 2026

Abstract

We give explicit presentation and oracle conventions for Poonen’s established transfer from rational polynomial solvability to rational-point existence on smooth projective geometrically integral varieties over $\mathbb{Q}$. A hypothetical decision procedure on this promised class would decide rational Hilbert’s tenth problem. A computable bound on the height of one rational point, when one exists, would likewise decide existence. These implications are conditional with respect to the proposed undecidability conclusion: the arithmetic input in OpenAI’s family 004 manuscript has not been certified by this audit. This research document makes no unconditional solution or publication claim.

1 Inputs and precise scope

Let $H10(\mathbb{Q})$ denote the decision problem with input an integer $n \geq 0$ and a polynomial $f \in \mathbb{Z}[x_1, \dots, x_n]$, asking whether $f(\mathbf{a}) = 0$ for some $\mathbf{a} \in \mathbb{Q}^n$. The number of variables is part of the input. Constants, including $f = 0$, are allowed.

A projective presentation e consists of an integer $N \geq 0$ and a finite list of homogeneous integer polynomials $F_1, \dots, F_r$ in $x_0, \dots, x_N$. It specifies

$$X_e = \text{Proj}(\mathbb{Q}[x_0, \dots, x_N]/(F_1, \dots, F_r)).$$

Define $\text{SPGI}(\mathbb{Q})$ to ask whether $X_e(\mathbb{Q}) \neq \emptyset$, under the promise that X_e is smooth and geometrically integral. Projectivity is supplied by the presentation. Geometric integrality includes nonemptiness of the geometric scheme. There is no bound on N , dimension, degrees, the number of equations, or coefficient heights. The oracle need not terminate on a presentation violating its promise.

The geometric construction and the decision transfer below are due to Poonen [1]. His Remark 1.2(c) credits a prior $\mathbb{Q}$ -case route to R. Robinson, through Smoryński, *Logical Number Theory I*, §II.7. We do not claim a new reduction. Family 004 [2] asserts the arithmetic undecidability input needed to turn this established implication into an unconditional geometric conclusion. Its validity is a separate issue; the statement appearing in a released manuscript is insufficient certification.

2 Poonen’s effective transfer and the promise

We use the following established input from Poonen’s Theorem 1.3: specialized to $\mathbb{Q}$: given a projective $\mathbb{Q}$ -variety X and an open $U \subseteq X$, one can algorithmically construct a regular projective variety Y with a morphism $\pi : Y \rightarrow X$ whose image on rational points is exactly $U(\mathbb{Q})$. The output need not be connected or geometrically integral.

The effectivity clause matters. In Poonen's §9, each permissible parameter c gives an output with $\pi_c(Y_c(\mathbb{Q})) \subseteq U(\mathbb{Q})$. An effective cardinality bound on the exceptional parameters allows a finite list containing a parameter for which equality holds. Taking the disjoint union of the outputs gives equality without identifying that good parameter. The boundary exclusion comes from his Châtelet-surface construction. Resolution preserves the lifts on the smooth fibers used in the argument. These are parts of the cited effective theorem, not a claim that naive projective closure preserves rational points.

Lemma 1. *If a connected smooth projective $\mathbb{Q}$ -scheme C has a rational point, then it is geometrically integral.*

Proof. A regular Noetherian scheme has disjoint irreducible components: at an intersection its local ring would have more than one minimal prime, whereas a regular local ring is a domain. Its finitely many components are therefore open and closed. Connectedness implies that C is integral. The finite algebra $L = H^0(C, \mathcal{O}_C)$ is consequently a field over $\mathbb{Q}$. A rational point induces a $\mathbb{Q}$ -algebra map $L \rightarrow \mathbb{Q}$, forcing $L = \mathbb{Q}$. Global functions commute with extension of the base field here. Over $\overline{\mathbb{Q}}$, each smooth connected projective component has only constant functions and is integral. Thus the number of these components is $\dim_{\mathbb{Q}} L = 1$. Smoothness also gives geometric reducedness, proving geometric integrality. This is the $\mathbb{Q}$ -projective instance of Poonen's Lemma 10.1. $\square$

The component decomposition and geometric-integrality tests used below are effective algebraic operations, as in Poonen's proof of Theorem 1.1(i). For clarity, for a smooth projective Y , effective coherent cohomology computes the finite *étale* algebra $H^0(Y, \mathcal{O}_Y)$. Its field factors give the connected components, and a connected component is geometrically integral precisely when its global-functions field has $\mathbb{Q}$ -dimension one. Projective constructions provide embeddings, and elimination gives homogeneous-equation presentations. Regularity and smoothness agree for finite-type schemes over the perfect field $\mathbb{Q}$. These conventions ensure that no invalid input is passed to the oracle.

Theorem 2. *An algorithm deciding $\text{SPGI}(\mathbb{Q})$ on every valid input would decide $\text{H10}(\mathbb{Q})$. Conversely, decidability of $\text{H10}(\mathbb{Q})$ would decide $\text{SPGI}(\mathbb{Q})$.*

Proof. For the forward implication let D be the promised decision procedure. On input f , handle constant polynomials directly. Otherwise form the reduced affine zero scheme U of f and its projective closure X , recording U as an open in X . If U is empty, return NO. Apply Poonen's effective theorem to obtain (Y, π) . Compute the connected components $C_1, \dots, C_s$ of Y and their projective presentations. Test geometric integrality and discard each component that fails. By Lemma 1, discarded components have no rational points. Every retained component is smooth, projective and geometrically integral. Query D on each retained presentation, and return YES if and only if some query answers YES.

Every preprocessing step terminates. The list of valid queries is finite, so D 's promise ensures termination of the whole algorithm. Correctness follows from

$$f(\mathbb{Q}^n) \ni 0 \iff U(\mathbb{Q}) \neq \emptyset \iff Y(\mathbb{Q}) \neq \emptyset \iff \exists i : C_i(\mathbb{Q}) \neq \emptyset.$$

The last condition is exactly the disjunction of the retained oracle answers. If there are no retained components it is false.

Conversely, on a projective presentation take the $N + 1$ standard affine charts, setting $x_j = 1$ on chart j . Their equations have a common rational zero precisely when the sum of their squares does, since $\mathbb{Q}$ is ordered. Call an $\text{H10}(\mathbb{Q})$ decision procedure on these finitely many polynomials and take the disjunction. Every rational projective point lies in one of these charts, proving the reverse implication. $\square$

The forward construction yields a finite nonadaptive disjunctive oracle reduction, and in particular a Turing reduction. It does not establish a single-output many-one reduction into the promised geometrically integral class. No quantitative or dimension-preservation assertion is used.

Corollary 3. *If $\text{H10}(\mathbb{Q})$ is undecidable, then no algorithm decides rational-point existence for the presented class $\text{SPGI}(\mathbb{Q})$.*

This corollary is conditional in the present audit. Its arithmetic antecedent must be verified independently before claiming that a hypothesis has been removed.

3 An explicit conditional height consequence

Specify a size parameter as follows. For $a \geq 0$, let $b(a)$ be its binary representation, with $b(0) = 0$, and let $\ell(a)$ be its length. Use the self-delimiting integer code $c(a) = 1^{\ell(a)}0b(a)$. Encode N, r , then the number of monomials in each polynomial. For every monomial encode a sign bit, c of the nonzero coefficient magnitude, and c of each of its $N + 1$ exponents. Require increasing lexicographic exponent vectors and a common total degree within each polynomial. The zero polynomial has zero monomials. These conventions specify a uniquely parsable finite string e ; let $s(e)$ be its bit length. The size refers to this presentation and embedding, not to the isomorphism class of the abstract variety.

For $P \in \mathbf{P}^N(\mathbb{Q})$ choose its primitive integer coordinates $(z_0, \dots, z_N)$ with the first nonzero coordinate positive, and set $H(P) = \max_j |z_j|$.

Corollary 4. *Assume $\text{H10}(\mathbb{Q})$ is undecidable. There is no total computable function $B : \mathbb{N} \rightarrow \mathbb{N}$ such that for every valid e with $X_e(\mathbb{Q}) \neq \emptyset$ there exists $P \in X_e(\mathbb{Q})$ satisfying $H(P) \leq B(s(e))$.*

Proof. Compute $M = B(s(e))$. Enumerate all nonzero primitive integer vectors in $[-M, M]^{N+1}$ with first nonzero coordinate positive, and evaluate the F_i exactly. Return YES if any vector satisfies every equation, and NO otherwise. The enumeration is finite. The hypothesized bound puts a point in the enumeration whenever one exists. This would decide $\text{SPGI}(\mathbb{Q})$, contradicting Corollary 3. For $M = 0$ the enumeration is empty, consistently with $H(P) \geq 1$. $\square$

The proof also rules out a presentation-dependent computable bound that terminates on every valid input. It concerns the height of *one* point when points exist. It makes no assertion about a fixed-dimensional family, finite rational-point sets, or all points on one variety.

4 Arithmetic validation gap and boundaries

Family 004's asserted main theorem has exactly the variable-count input convention above. Its proof uses finite rational tests, compactness, additive elliptic indices and arithmetic height estimates. The current audit has reconstructed the conditional transfer and several arithmetic interfaces, but has not certified all pivotal new companion arguments. No applicable complete Lean formalization of family 004 has been identified; this is a verification limit, not evidence that the theorem is false.

One particularly precise remaining dependency occurs in its pole-parity argument. For the special curves

$$E_l : y^2 = x(x - l)(x + 3l),$$

with l restricted to the parameters selected by family 004’s descent construction, that construction gives $\dim_{\mathbf{F}_2} \text{Sel}_2(E_l/\mathbb{Q}) = 3$ and full rational two-torsion. Hence

$$r + \dim_{\mathbf{F}_2} \text{Sha}(E_l/\mathbb{Q})[2] = 1, \quad r = \text{rank } E_l(\mathbb{Q}).$$

The desired realization of its extra Selmer class by a rational point follows if $\text{Sha}(E_l/\mathbb{Q})[2^\infty]$ is finite: the alternating perfect Cassels–Tate pairing then forces even dimension of $\text{Sha}[2]$, yielding $r = 1$ and $\text{Sha}[2] = 0$. Low Selmer corank alone does not supply that finiteness. The cited new pointwise 2-converse [3] is the upstream assertion intended to do so and remains under audit. This document does not assume its conclusion by silently replacing it with a named conjecture or a mismatched formalization.

Naive compactification cannot replace the effective transfer: $\text{Spec } \mathbb{Q}[x, y]/(x^2 + 1)$ has no rational point, whereas its closure $V(X^2 + Z^2) \subseteq \mathbf{P}^2$ contains $[0 : 1 : 0]$. Resolution alone need not preserve point existence in the other direction: $V(x^2 + y^2)$ has a rational singular point but its normalization is $\mathbf{A}_{\mathbb{Q}(i)}^1$, which has no rational point. A morphism from a resolution cannot create a rational point when its target has none. The issue is losing singular points or retaining boundary points introduced by compactification.

No conclusion for curves, surfaces, fixed dimension, Fano varieties, general type, or a finite-point promise follows automatically. This audit asserts neither a Turing degree nor quantitative bounds beyond the stated oracle implication. The height corollary is an elementary computability consequence and adds no independent arithmetic input.

Disclosure and status. AI tools were used extensively in research, drafting and adversarial verification. Those automated reviews are not conventional human peer review. This is a local conditional research document; it has not been deposited as an unconditional solution or conventionally refereed.

References

- [1] Bjorn Poonen, *Existence of rational points on smooth projective varieties*, J. Eur. Math. Soc. **11** (2009), 529–543. [doi:10.4171/JEMS/159](https://doi.org/10.4171/JEMS/159).
- [2] OpenAI, *Hilbert’s tenth problem over the rational numbers*, OpenAI Math Release preprint, September 24, 2026. Manuscript key: OAI:Hilberts-tenth-problem-over-the-rational-numbers-September-24-2026. Source version: commit `adc7f1241b42e322a6451854ab7e4b4c146bf78a`. [Pinned source and citation metadata](#).
- [3] OpenAI, *A pointwise 2-converse for elliptic curves with rational two-torsion*, OpenAI Math Release preprint, September 24, 2026, same pinned commit. [Pinned source and citation metadata](#).